

DATA ANALYSIS COURSE
DIAGNOSTIC ANALYSIS

SHOPEE BUSINESS INTELLIGENCE

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BUSINESS CONTEXT & PROBLEM DEFINITION

OVERVIEW

CONTEXT

There is currently a promotion offering 30% off the total bill, capped at 20.000. This applies to all transactions.

PROBLEM

There are signs of collusion between buyers and sellers to abuse the promotion by creating multiple orders to maximize the rebate

QUESTIONS

- Which total bill range has the highest number of transactions?
- Who are the top 3 sellers most likely to be committing fraud, and why?
- Who are the top 3 buyers most likely to be committing fraud, and why?



HYPOTHESIS

Maximum discount condition:

- $30\% \times \text{GMV} = 20,000$
→ $\text{GMV} \approx 66,667$

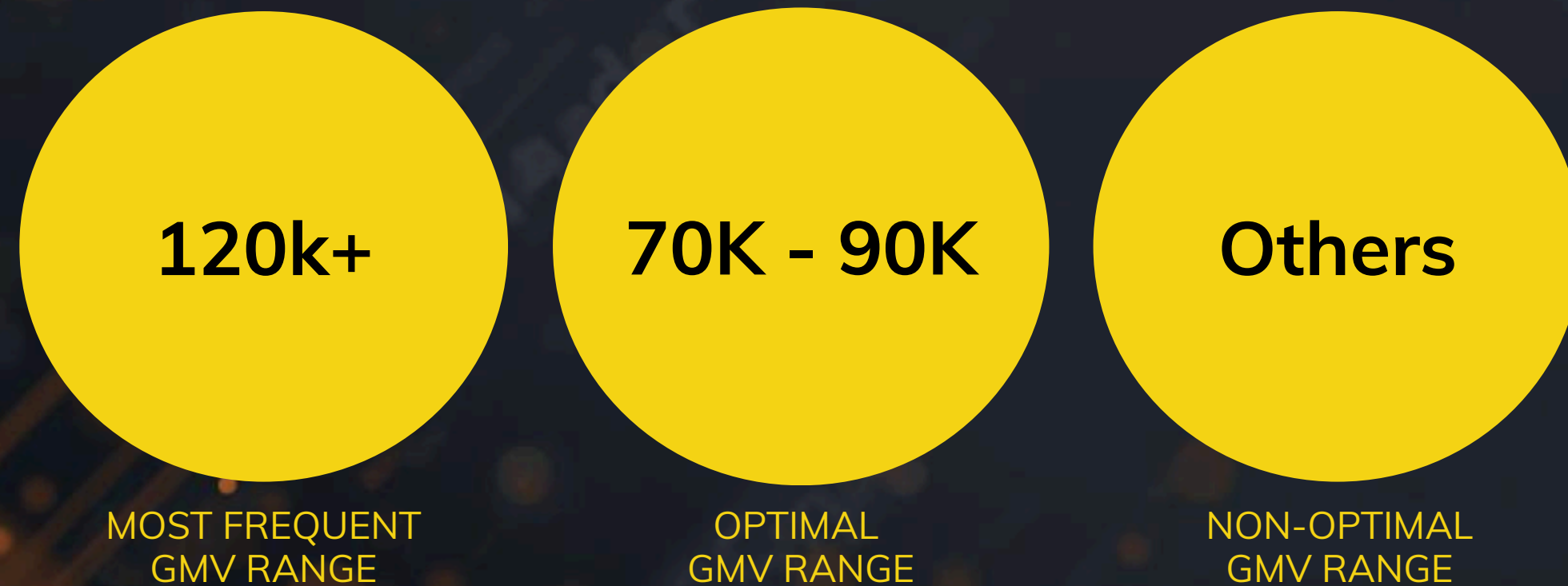
Expected behavior if abuse exists:

- Transactions concentrate around 70.000 - 90.000
- Rebate frequently reaches maximum (20,000)
- Repeated transactions from the same users and the same shops

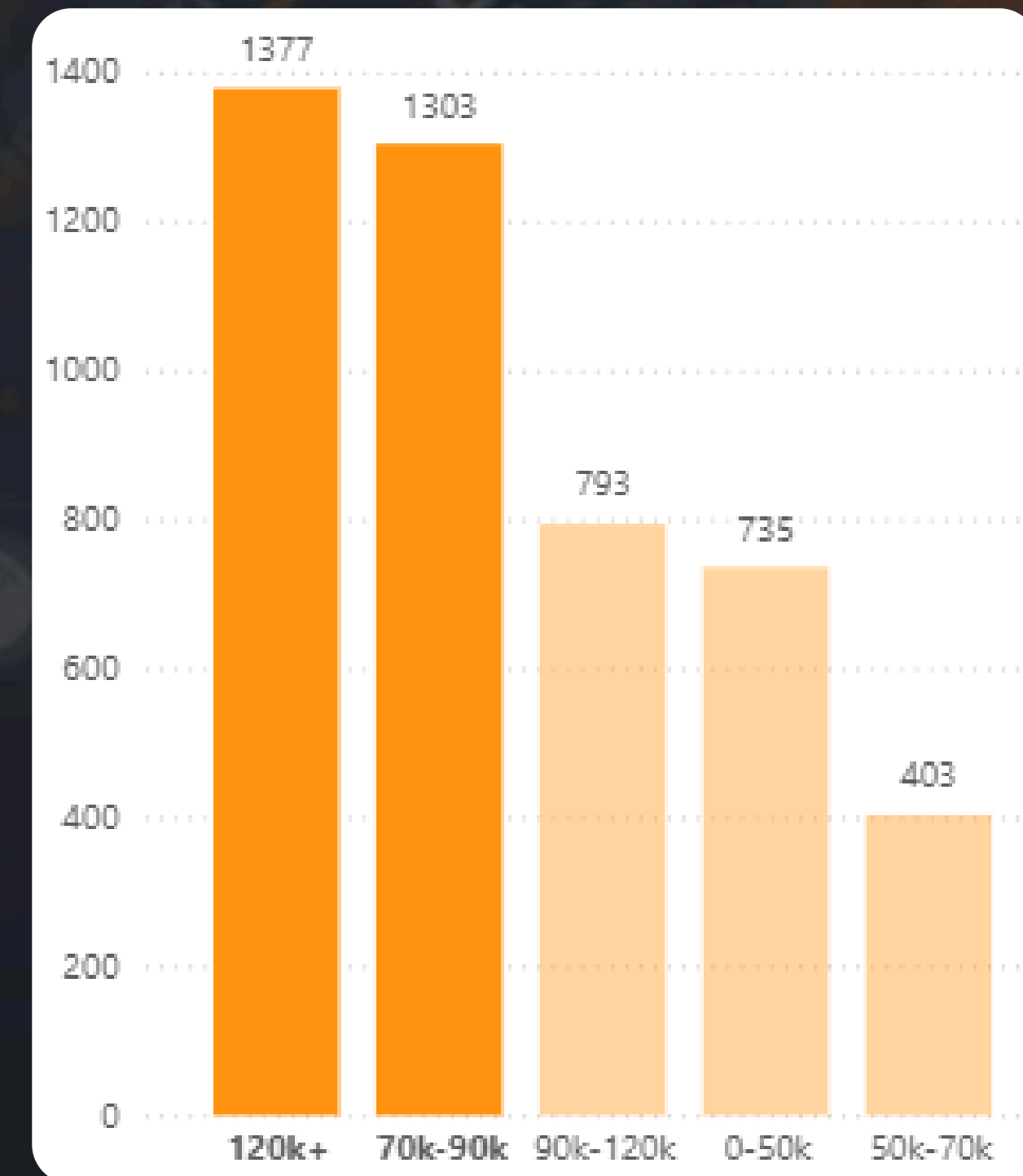
==> Users may intentionally optimize order values to maximize the promotion benefit



GMV DISTRIBUTION



- Transactions are heavily concentrated in the **70k–90k range** with total **1303** transactions, which aligns closely with the optimal value to maximize the 20,000 discount.
- This pattern suggests non-random behavior and indicates potential promotion abuse.



TOP 3 SUSPICIOUS SELLERS

KEY FINDINGS

- Top 3 Suspicious Sellers can be spotted in the table, they are shops with id: 4809, 5786, 5808
- These sellers have a high number of transactions with GMV consistently in the optimal range and nearly 100% rebate usage.
 - This pattern strongly suggests intentional behavior to maximize the promotion benefit.

FRAUD INDICATORS

- Repeated transactions with maximum rebate
- Order values optimized to maximize discount
- Behavior is systematic, not random

shop_id	avg_gmv	total_orders	rebate_ratio
4809	82,548.39	31	1
5786	86,966.67	30	1
5808	84,206.90	29	1
7873	79,130.43	23	1
9222	79,304.35	23	1
8224	80,565.22	23	1
4820	81,052.63	19	1
7702	88,210.53	19	1
6917	89,642.86	14	1
29499	71,615.38	13	1
6840	85,615.38	13	1
6909	88,692.31	13	1
9589	76,166.67	12	1
6809	81,181.82	11	1
7907	83,636.36	11	1
9096	85,363.64	11	1
1782	70,555.56	9	1
3230	71,111.11	9	1
5681	73,888.89	9	1
10594	77,777.78	9	1
5055	85,000.00	8	1

TOP 3 SUSPICIOUS BUYERS

KEY FINDINGS

- Top 3 Suspicious Buyers can be spotted in the table, they are users with id: **103023839**, **103366324**, **103653018**
- These buyers consistently make transactions within the optimal GMV range and receive the maximum rebate almost every time
- This behavior indicates potential abuse of the promotion
- Compared to normal users, these buyers have significantly higher rebate ratios and more concentrated transaction values.

FRAUD INDICATORS

- Repeated purchases at optimal discount threshold
- Consistently receiving maximum rebate (20,000 VND)
- Unusual transaction pattern compared to normal users

uid	avg_gmv	total_orders	rebate_ratio
103023839	88,200.00	10	1
103366324	83,000.00	10	1
103653018	87,000.00	10	1
103653392	85,000.00	10	1
103667901	86,000.00	10	1
103682325	80,000.00	10	1
103002018	70,555.56	9	1
103178855	71,111.11	9	1
103266260	78,333.33	9	1
103855768	76,111.11	9	1
100702363	70,000.00	8	1
103044301	70,250.00	8	1
103044781	72,500.00	8	1
103033590	70,714.29	7	1
103044677	72,142.86	7	1
103690355	78,571.43	7	1
101166204	71,666.67	6	1
101793553	70,000.00	6	1
103333474	76,666.67	6	1
103481792	87,333.33	6	1
103403300	85,333.33	5	1

BUYER-SELLER COLLUSION

KEY FINDINGS

- We observe that certain buyer-seller pairs **repeatedly transact with each other, consistently receiving the maximum rebate.**
- These repeated interactions between specific buyers and sellers strongly suggest coordinated behavior to exploit the promotion

FRAUD INDICATORS

- Same buyer and seller repeatedly transact
- Transactions consistently fall in optimal GMV range
- High frequency + max rebate → strong anomaly

shop_id	uid	Count of order_id
2144	104029244	4
4673	1225554	4
5417	103381398	4
5643	100702363	4
5681	100869997	4
7747	103271941	4
9096	100451463	4
9524	103286330	4
10979	1638171	4
10979	1877192	4
10979	101785843	4
10979	103175209	4
2613	2029746	3
3760	100256932	3
5681	103975257	3
6659	1924682	3
6659	101166204	3
6732	101166204	3
6732	101476221	3
6732	103160000	3
Total		1264

CONCLUSION & RECOMMENDATION

Key Conclusions

- Transactions are heavily concentrated in the optimal GMV range (70k–90k)
- Both sellers and buyers show:
 - High transaction frequency
 - Consistent maximum rebate usage
- Specific buyer–seller pairs exhibit:
 - Repeated interactions
 - Strong signs of collusion

Recommendations

1. Limit Promotion Abuse

- Set daily/weekly usage limits per user
- Restrict number of discounted transactions

2. Fraud Detection Rules

- Flag users with:
 - High rebate ratio (>90%)
 - GMV consistently in optimal range
- Monitor repeated buyer–seller pairs

3. System Improvements

- Introduce anomaly detection models
- Real-time monitoring of suspicious patterns

Business Impact

- Reduce financial loss from promotion abuse
- Ensure fairness for legitimate users
- Improve campaign effectiveness





QUESTIONS & FEEDBACK

THANK YOU